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Integrative Transcriptomic and Network Analysis of Somatic Cell Reprogramming Toward Pluripotency Highlights SPRY2 as a Candidate Gene

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Abstract

This study presents an integrative bioinformatic re-analysis of publicly available transcriptomic data to characterize the transcriptional and functional changes underlying somatic cell reprogramming toward pluripotency and to identify candidate genes potentially associated with this process. RNA-sequencing data from the Gene Expression Omnibus SubSeries GSE90894, part of the GSE90895 SuperSeries, were analyzed in the R environment. The dataset comprised *Mus musculus* cells representing the reprogramming trajectory from mouse embryonic fibroblasts (MEFs) through 48 h of OSKM induction and pre-iPSC states to embryonic stem cells (ESCs). Following preprocessing and quality control, differential expression and functional enrichment analyses were performed using limma with Benjamini–Hochberg correction, together with Gene Ontology, KEGG, Reactome, and gene set enrichment analysis (GSEA). The results demonstrated progressive transcriptional remodeling during reprogramming, characterized by suppression of extracellular matrix, collagen, adhesion, cytoskeletal, and epithelial–mesenchymal transition-associated programs. This was accompanied by enrichment of RNA-related and cell-cycle programs at intermediate stages and pluripotency-, germ-cell-, and epigenetic-associated programs at later stages. No genes met the stringent criteria of $FDR < 0.05$ and $|\log_2FC| \geq 1$ in the analyzed contrasts. In the ESC–MEF comparison, 2348 genes met the exploratory criteria of nominal $P < 0.05$ and

$|\log_2\text{FC}| \geq 1$. SPRY2 demonstrated a consistent positive expression trend across the reprogramming trajectory, with the largest increase in expression observed during the transition from 48 h to the pre-iPSC stage and an overall $\log_2\text{FC}$ of +1.23 in ESCs versus MEFs, although this change was not statistically significant after multiple-testing correction. Network analysis revealed distinct pluripotency-associated and extracellular matrix-associated gene clusters, while SPRY2 did not show evident connections within the selected STRING network and was retained as an exploratory candidate. These findings provide an integrative view of transcriptional remodeling during somatic cell reprogramming and highlight SPRY2 as a candidate for further investigation of the regulatory mechanisms underlying cellular plasticity and acquisition of pluripotency.

Keywords

Somatic cell reprogramming; pluripotency; transcriptomic analysis; bioinformatics; RNA-seq; differential gene expression; functional enrichment; SPRY2; cellular plasticity; *Mus musculus*.

Introduction

Somatic cell reprogramming toward pluripotency involves extensive remodeling of cellular identity and gene expression, enabling differentiated cells to acquire characteristics of a pluripotent state. This process is driven by coordinated transcriptional and regulatory changes involving the core reprogramming factors OCT4 (POU5F1), SOX2, KLF4, and MYC, together with changes in signaling, cell-cycle regulation, metabolism, and epigenetic state. High-throughput transcriptomic approaches enable characterization of these molecular changes across successive stages of reprogramming. Public repositories such as the Gene Expression Omnibus (GEO) [1] provide valuable datasets for reproducible secondary analyses aimed at identifying differentially expressed genes, altered biological processes, and molecular networks associated with changes in cellular identity.

SPRY2 (Sprouty homolog 2) is an intracellular regulator of receptor tyrosine kinase signaling that modulates the RAS–RAF–MEK–ERK signaling axis [4]. Its effects on ERK signaling can depend on cellular and signaling context, while its involvement in growth-factor signaling and cellular responses makes SPRY2 a biologically relevant candidate for investigation in the context of cellular plasticity. However, its transcriptional dynamics during the transition from a somatic to a pluripotent state remain insufficiently characterized during the analyzed reprogramming trajectory [7].

Therefore, this study aimed to characterize transcriptional and functional changes across the trajectory of somatic cell reprogramming toward pluripotency using integrative bioinformatic and network analysis of publicly available RNA-seq data, and to investigate SPRY2 as an exploratory candidate associated with cellular plasticity and the acquisition of pluripotency.

Materials and Methods

Transcriptomic dataset and study design

Publicly available RNA-seq data were obtained from the Gene Expression Omnibus (GEO) database [1]. The analysis was based on SubSeries GSE90894 within the GSE90895 SuperSeries, comprising *Mus musculus* transcriptomic profiles obtained during somatic cell reprogramming. The primary analysis included samples representing mouse embryonic fibroblasts (MEFs), 48 h after OSKM induction, pre-iPSC cells, and embryonic stem cells (ESCs), defining the trajectory MEFs → 48 h → pre-iPSC → ESCs. Four pairwise contrasts were analyzed: 48 h vs MEFs, pre-iPSC vs 48 h, ESC vs pre-iPSC, and ESC vs MEFs.

Data preprocessing and quality control

The supplementary expression matrix GSE90894_RPKM_mRNAseq_table.xlsx was imported into R. Gene identifiers were assigned as row names, and duplicate identifiers were resolved using `make.unique()` without removing expression measurements. The

resulting matrix contained 22,608 genes across 11 samples. Expression values were transformed using $\log_2(\text{RPKM} + 1)$, followed by removal of genes with zero variance, yielding a filtered matrix of 19,801 genes. Data quality and sample relationships were assessed using boxplots, principal component analysis (PCA), pairwise correlation analysis, and hierarchical clustering.

Differential expression analysis

Differential expression analysis was performed using the limma framework. A design matrix and four corresponding contrasts were constructed for the predefined reprogramming stages [5]. Linear models were fitted and moderated using the empirical Bayes method implemented in eBayes.

For each gene, $\log_2\text{FC}$, nominal P -value, and Benjamini–Hochberg-adjusted P -value (FDR) were calculated. Genes with $\text{FDR} < 0.05$ and $|\log_2\text{FC}| \geq 1$ were considered statistically significant DEGs. Because no genes met these criteria, exploratory gene sets were additionally defined using nominal $P < 0.05$ and $|\log_2\text{FC}| \geq 1$ and were interpreted separately from statistically significant DEGs.

Functional enrichment and GSEA

Functional enrichment was assessed using GO Biological Process [2], KEGG [3], and Reactome analyses with *Mus musculus* annotation from org.Mm.eg.db. Exploratory upregulated and downregulated gene sets were analyzed separately for the major contrasts. Multiple testing was controlled using the Benjamini–Hochberg procedure. GSEA [6] was additionally performed using ranked differential-expression statistics to assess pathway-level changes across the transcriptome without applying a predefined gene-level significance threshold.

SPRY2 expression analysis

Spry2 expression was examined across the four predefined contrasts. Its $\log_2\text{FC}$, nominal P -value, and FDR were extracted from the corresponding limma results. SPRY2 was

treated as an exploratory candidate because none of its expression changes reached the predefined FDR threshold.

Network analysis

For network analysis, the 50 genes with the strongest positive and 50 genes with the strongest negative moderated *t*-statistics in the ESC–MEF comparison were selected. SPRY2 was added as a candidate gene, resulting in a 101-gene set. Protein–protein interaction relationships were assessed using STRING [7]. The resulting network was examined for functional associations among genes related to pluripotency and extracellular matrix-associated programs, with SPRY2 evaluated as an exploratory candidate.

SPRY2 correlation and functional analysis

Pearson correlation coefficients between SPRY2 expression and the expression of all analyzed genes were calculated across the 11 samples. The 30 strongest positive and 30 strongest negative correlations were selected for exploratory functional analysis [9].

The resulting 60-gene set was subjected to GO Biological Process enrichment analysis. For Reactome analysis, gene symbols were converted to Entrez Gene IDs using org.Mm.eg.db; unmapped and duplicate identifiers were excluded. Reactome enrichment was subsequently performed using the mouse pathway database with multiple-testing correction.

Data and code availability

All scripts used for data preprocessing, statistical analysis, visualization, and generation of the reported figures are available in the project GitHub repository: <https://github.com/AlinaSobolieva/Integrative-Analysis-of-Somatic-Cell-Reprogramming-Identifies-SPRY2-as-a-Candidate>

Results

To characterize transcriptional changes associated with somatic cell reprogramming, the complete set of 11 RNA-seq samples was first evaluated for expression quality and sample relationships, whereas subsequent stage-specific analyses focused on the MEF → 48 h → pre-iPSC → ESC trajectory. Following $\log_2$ transformation, expression distributions became broadly comparable across samples, providing a consistent scale for subsequent analyses. PCA revealed separation of the major cellular states, with ESCs clearly distinct from MEF-derived samples along PC1. MEF and 48 h samples occupied relatively close positions, whereas the two pre-iPSC samples showed greater dispersion: pre-iPSC #1 was positioned near the 48 h samples, while pre-iPSC #2 was separated along PC2. PC1 and PC2 accounted for 32.5% and 17.9% of the total variance, respectively, together explaining 50.4% of the observed variation. Pairwise Pearson correlations were generally high ($r = 0.79\text{--}1.00$), indicating substantial overall similarity among the transcriptomic profiles. MEF and 48 h samples showed particularly high within-group similarity, with correlations predominantly above 0.93. ESCs exhibited lower correlations with MEF-derived samples ($r = 0.79\text{--}0.86$), consistent with their distinct transcriptional state. Despite their separation in PCA space, the two pre-iPSC samples showed a high mutual correlation ($r = 0.93$). Hierarchical clustering provided a complementary assessment of sample relationships and is presented in Supplementary Figure 1.

Overall, the analyses demonstrated structured transcriptional variation across reprogramming stages, with pronounced separation of ESCs and increased heterogeneity at the pre-iPSC stage.

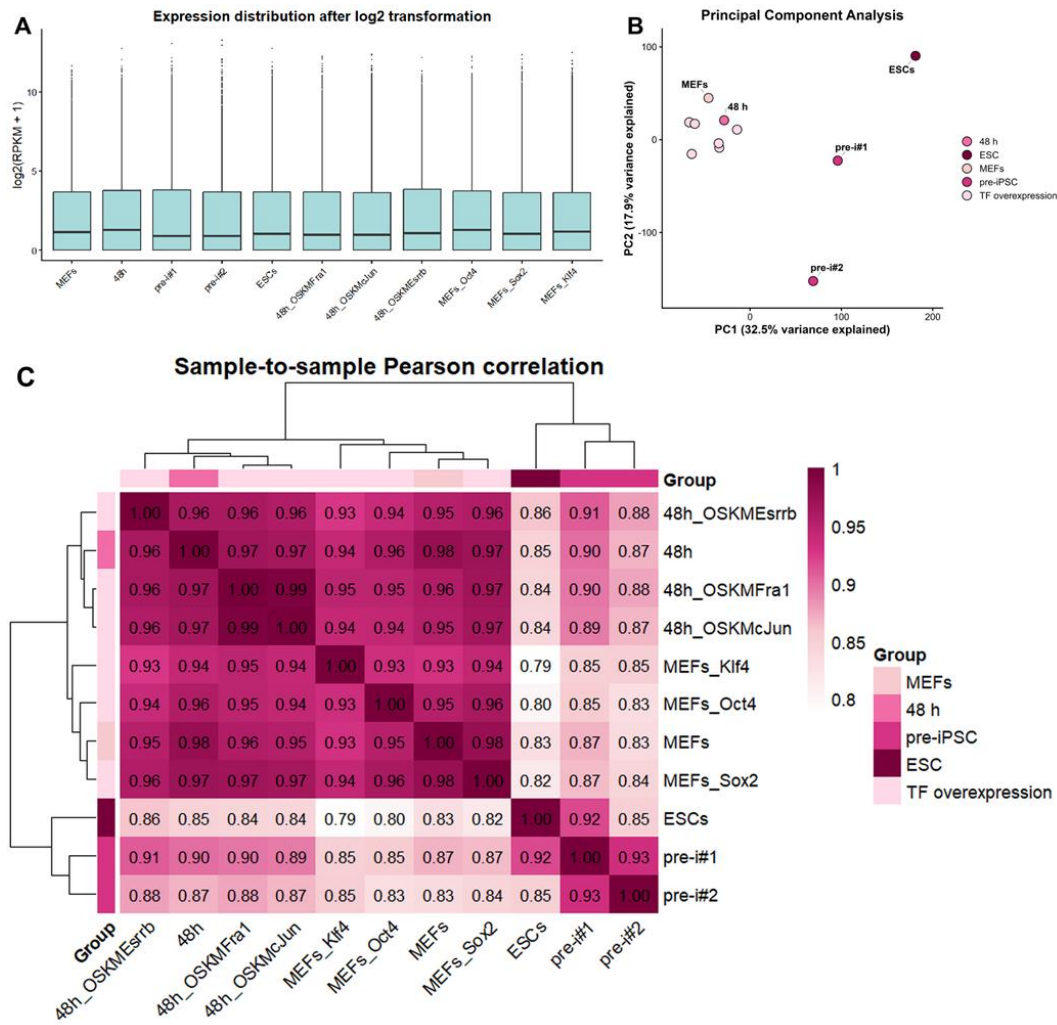


Figure 1. Transcriptomic data quality and sample relationships. (A) Distribution of $\log_2(\text{RPKM} + 1)$ expression values across the analyzed samples. (B) Principal component analysis of the analyzed samples. (C) Pairwise Pearson correlation matrix showing sample-to-sample expression similarity.

Differential expression analysis across the four predefined contrasts revealed extensive transcriptional remodeling, although no genes met the stringent criteria of $\text{FDR} < 0.05$ and $|\log_2\text{FC}| \geq 1$. Exploratory analysis using nominal $P < 0.05$ and $|\log_2\text{FC}| \geq 1$ identified 340 genes in 48 h vs MEF, 1436 in pre-iPSC vs 48 h, 1304 in ESC vs pre-iPSC, and 2348 in ESC vs MEF (Figure 2). These genes were considered exploratory candidates rather than statistically validated DEGs. The largest transcriptional shift was observed between ESCs and MEFs, reflecting the transition from the initial somatic to the final pluripotent state.

Genes associated with pluripotency and ESC identity, including Pou5f1, Esrrb, Lin28a, and Epcam, showed positive expression changes, whereas genes associated with extracellular matrix and mesenchymal programs were predominantly reduced. SPRY2 showed a consistent positive expression trend across all four contrasts, with $\log_2\text{FC}$ values of +0.318, +0.702, +0.210, and +1.230, respectively. The largest stage-specific increase occurred at the 48 h $\rightarrow$ pre-iPSC transition, while the strongest overall difference was observed between ESCs and MEFs. However, none of these changes reached statistical significance after multiple-testing correction, supporting the classification of SPRY2 as an exploratory candidate rather than a significant DEG.

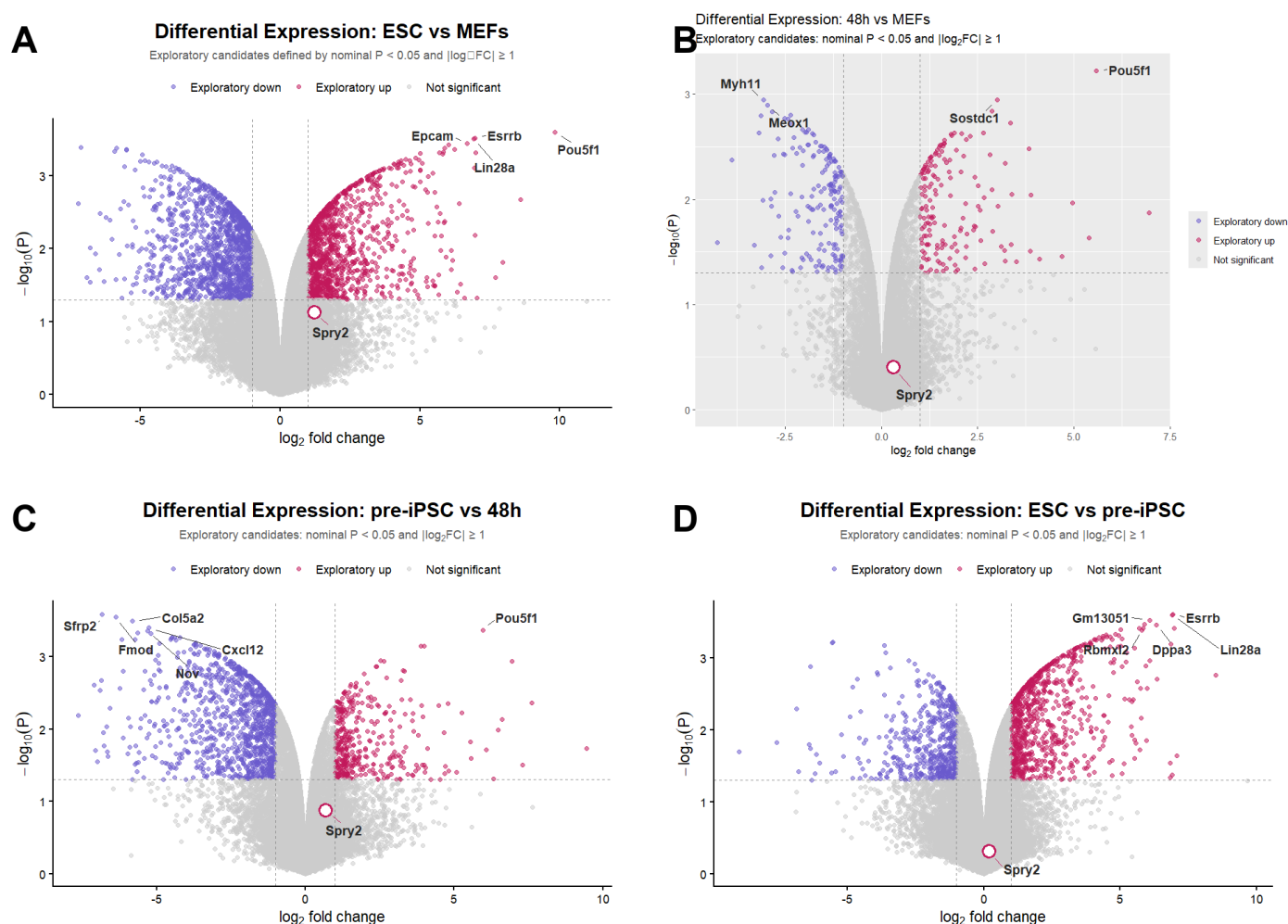


Figure 2. Differential expression across the reprogramming trajectory. Volcano plots show (A) ESC vs MEF, (B) 48 h vs MEF, (C) pre-iPSC vs 48 h, and (D) ESC vs pre-iPSC.

Dashed lines indicate the exploratory thresholds of nominal $P < 0.05$ and $|\log_2FC| \geq 1$. Additional enrichment results across the analyzed stages are provided in Supplementary Figure 2.

Functional enrichment analysis revealed stage-specific remodeling of biological programs during the transition from MEFs toward the ESC state (Figure 3). At the early stage, downregulated genes were associated with extracellular matrix, cell adhesion, cytoskeletal, and mesenchymal-related processes, indicating an early suppression of the MEF-associated program. At the intermediate stage, these changes became more pronounced, with downregulation of ECM, collagen, focal adhesion, integrin, and cytoskeletal programs and enrichment of RNA- and ribosome-associated processes among upregulated genes. GSEA additionally showed enrichment of MYC targets, E2F targets, and G2M checkpoint, indicating active transcriptional and cell-cycle remodeling. At the late stage, upregulated genes were enriched in germ-cell development, meiosis, and pluripotency-associated pathways, while downregulated genes were associated with immune and signaling processes. In the ESC–MEF comparison, this transition was most pronounced, with strong depletion of ECM-, collagen-, adhesion-, and mesenchymal/EMT-related programs. GSEA confirmed suppression of EMT (NES = -2.26 , FDR = 5×10^{-9}). Gene–concept network analysis identified functionally connected groups of genes associated with the enriched biological processes. Among the most highly connected genes, SPRY2 was associated with ERK1/ERK2 signaling, phosphorylation, and epithelial differentiation, together with DUSP6, AREG, CLSPN, CD24, NT5E, and CD36.

Overall, the enrichment analyses indicate a progressive loss of the MEF-associated mesenchymal/ECM program accompanied by acquisition of transcriptional programs characteristic of the ESC state.

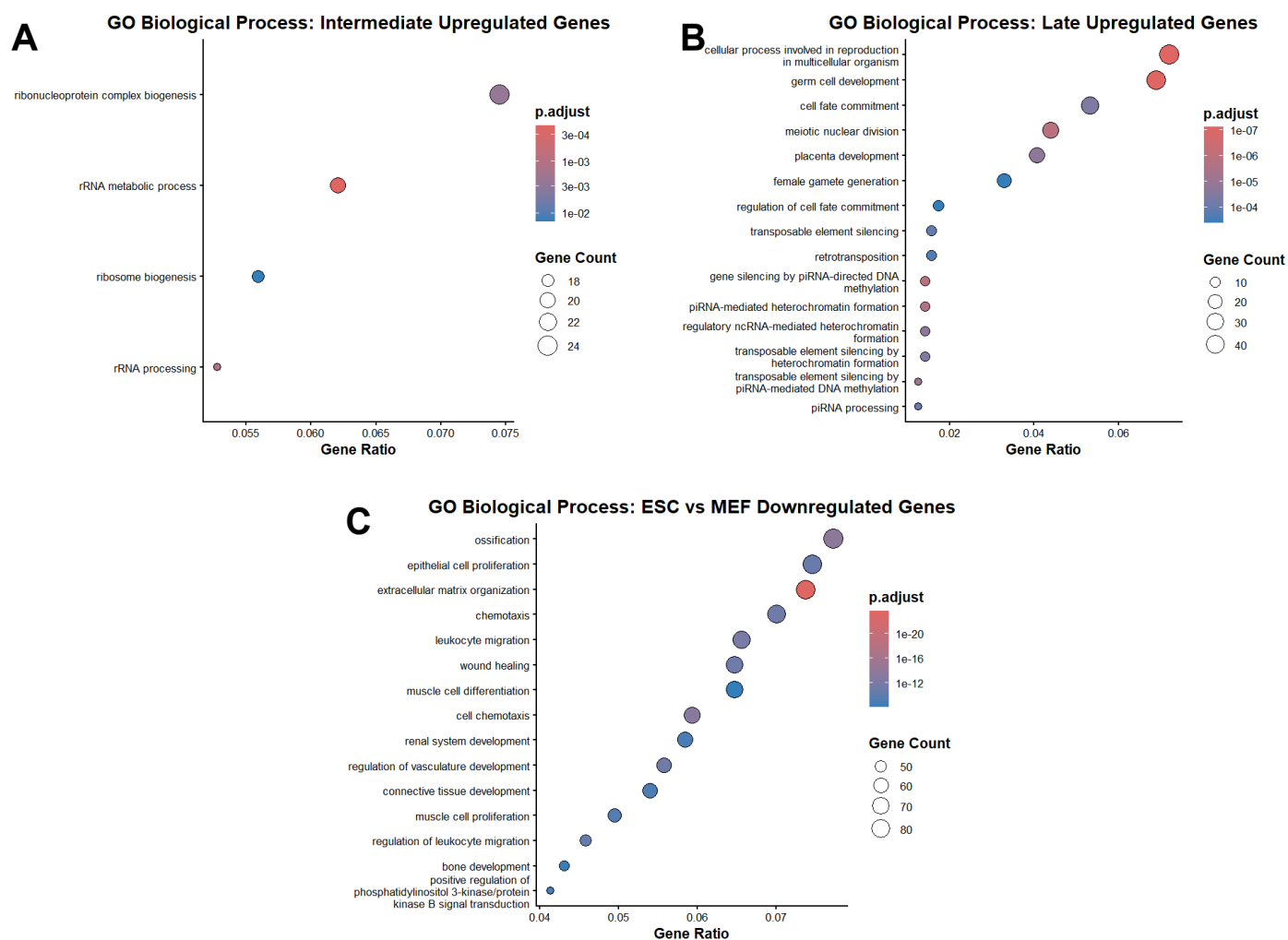


Figure 3. Functional remodeling during somatic cell reprogramming. Representative enrichment results are shown for (A) intermediate-stage upregulated genes, (B) late-stage upregulated genes, and (C) genes downregulated in ESCs relative to MEFs.

Given its positive expression trend across the reprogramming trajectory, SPRY2 was examined as an exploratory candidate (Figure 4). It showed positive shifts across all four contrasts, with the largest shift at 48 h → pre-iPSC and the strongest overall difference in ESC–MEF ($\log_2\text{FC} = +1.23$), although this effect was not statistically significant ($\text{FDR} = 0.314$). Pearson correlation analysis identified genes strongly co-varying with SPRY2 across the 11 samples. Exploratory enrichment of the 60 strongest positive and negative correlations revealed significant Reactome enrichment for transport of mature mRNAs derived from intronless transcripts, involving Nup93, Nup62, and Cpsf4 (FDR

= 0.011), whereas GO enrichment was not significant after multiple-testing correction. SPRY2 was therefore retained as an exploratory candidate [2], while its expression pattern did not provide evidence for statistically significant differential expression or causality. STRING analysis further supported the interaction context of SPRY2, showing associations with signaling-related proteins including EGFR, GRB2, BRAF, and DUSP6 (Supplementary Figure S3).

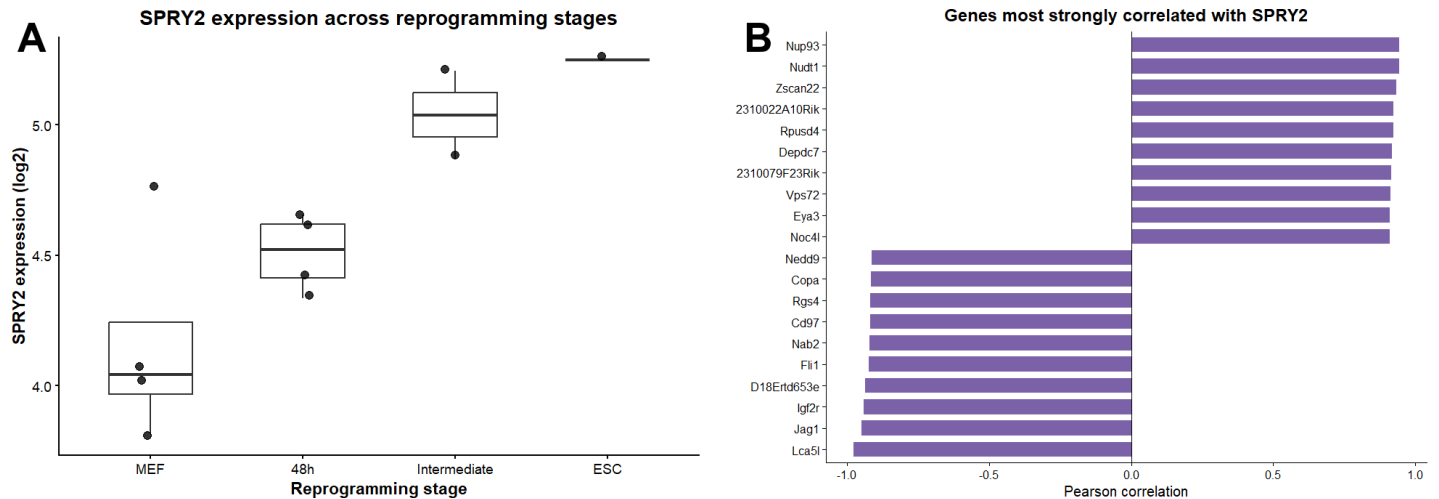


Figure 4. SPRY2 expression and transcriptional context during somatic cell reprogramming. (A) SPRY2 expression across the major reprogramming stages. (B) Genes showing the strongest positive and negative correlations with SPRY2 expression across the analyzed samples.

Discussion

Transcriptional remodeling during somatic cell reprogramming

The present analysis revealed progressive transcriptional remodeling during the transition from MEFs toward an ESC state. The dominant pattern was suppression of ECM, collagen, adhesion, cytoskeletal, and EMT-associated programs, accompanied by enrichment of RNA-related, cell-cycle, pluripotency-associated, and germ-cell-related programs. The intermediate stage showed particularly pronounced remodeling, combining repression of mesenchymal programs with enrichment of MYC, E2F, and G2M programs.

At later stages, pluripotency- and germ-cell-associated programs became more prominent, consistent with broader developmental and epigenetic remodeling accompanying acquisition of an ESC-like state. Importantly, germ-cell and spermatogenesis enrichment should not be interpreted as evidence of germ-cell identity.

SPRY2 as an exploratory candidate

SPRY2 showed a consistent positive expression trend across the reprogramming trajectory, with the strongest overall shift observed between ESCs and MEFs ($\log_2FC = +1.23$). However, this change did not reach statistical significance after multiple-testing correction ($FDR = 0.314$). Thus, SPRY2 cannot be considered a statistically validated DEG based on the present dataset. Nevertheless, its reproducible positive trend across the analyzed contrasts supported its prioritization for exploratory network and correlation analyses. SPRY2-associated genes showed enrichment of mRNA transport processes involving intronless transcripts, although GO enrichment did not remain significant after correction. These findings provide a transcriptional context for SPRY2 but do not establish a direct regulatory or causal role in reprogramming [4].

Limitations and implications

The main limitations are the small sample size, limited statistical power, and absence of independent experimental validation. Nevertheless, the consistent transcriptional remodeling observed across stages supports SPRY2 as an exploratory candidate for further investigation in cellular reprogramming.

Conclusions

The integrative analysis revealed stage-specific transcriptional remodeling during somatic cell reprogramming toward pluripotency, characterized by progressive suppression of mesenchymal/ECM-associated programs and emergence of pluripotency- and cell-state-associated transcriptional programs. SPRY2 showed a consistent positive expression trend

and was retained as an exploratory candidate for further investigation, although its differential expression was not statistically significant after multiple-testing correction.

Conflict of Interest

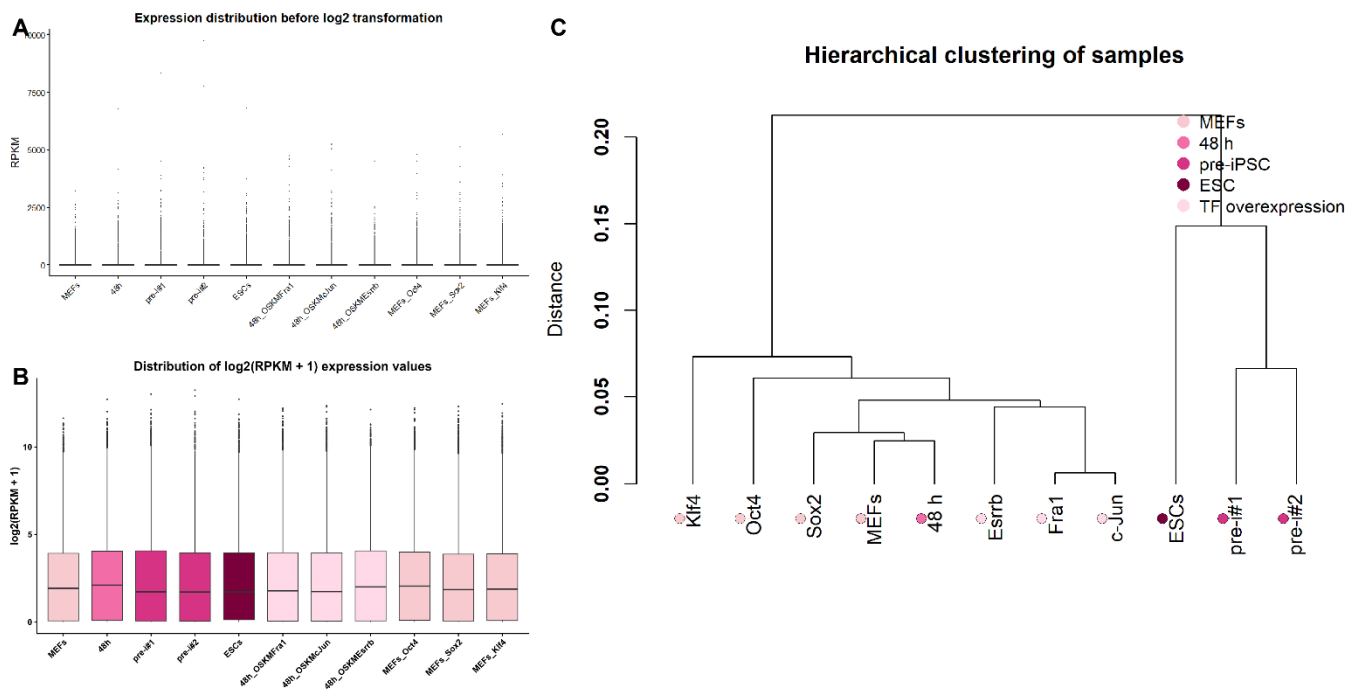
The authors declare no conflict of interest.

References

1. Edgar R, Domrachev M, Lash AE. Gene Expression Omnibus: NCBI gene expression and hybridization array data repository. *Nucleic Acids Research*. 2002;30(1):207–210. doi:10.1093/nar/30.1.207.
2. Gene Ontology Consortium. The Gene Ontology resource: enriching a GOld mine. *Nucleic Acids Research*. 2021;49(D1):D325–D334. doi:10.1093/nar/gkaa1113.
3. Kanehisa M, Goto S. KEGG: Kyoto Encyclopedia of Genes and Genomes. *Nucleic Acids Research*. 2000;28(1):27–30. doi:10.1093/nar/28.1.27.
4. Lao DH, Chandramouli S, Yusoff P, Fong CW, Saw TY, Tai LP, Yu CY, Leong HF, Guy GR. A Src homology 3-binding sequence on the C terminus of Sprouty2 is necessary for inhibition of the Ras/ERK pathway downstream of fibroblast growth factor receptor stimulation. *Journal of Biological Chemistry*. 2006;281(40):29993–30000. doi:10.1074/jbc.M604044200.
5. Ritchie ME, Phipson B, Wu D, Hu Y, Law CW, Shi W, Smyth GK. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*. 2015;43(7):e47. doi:10.1093/nar/gkv007.
6. Subramanian A, Tamayo P, Mootha VK, Mukherjee S, Ebert BL, Gillette MA, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences of the United States of America*. 2005;102(43):15545–15550. doi:10.1073/pnas.0506580102.
7. Szklarczyk D, Kirsch R, Koutrouli M, Nastou K, Mehryary F, Hachilif R, et al. The STRING database in 2023: protein–protein association networks and functional

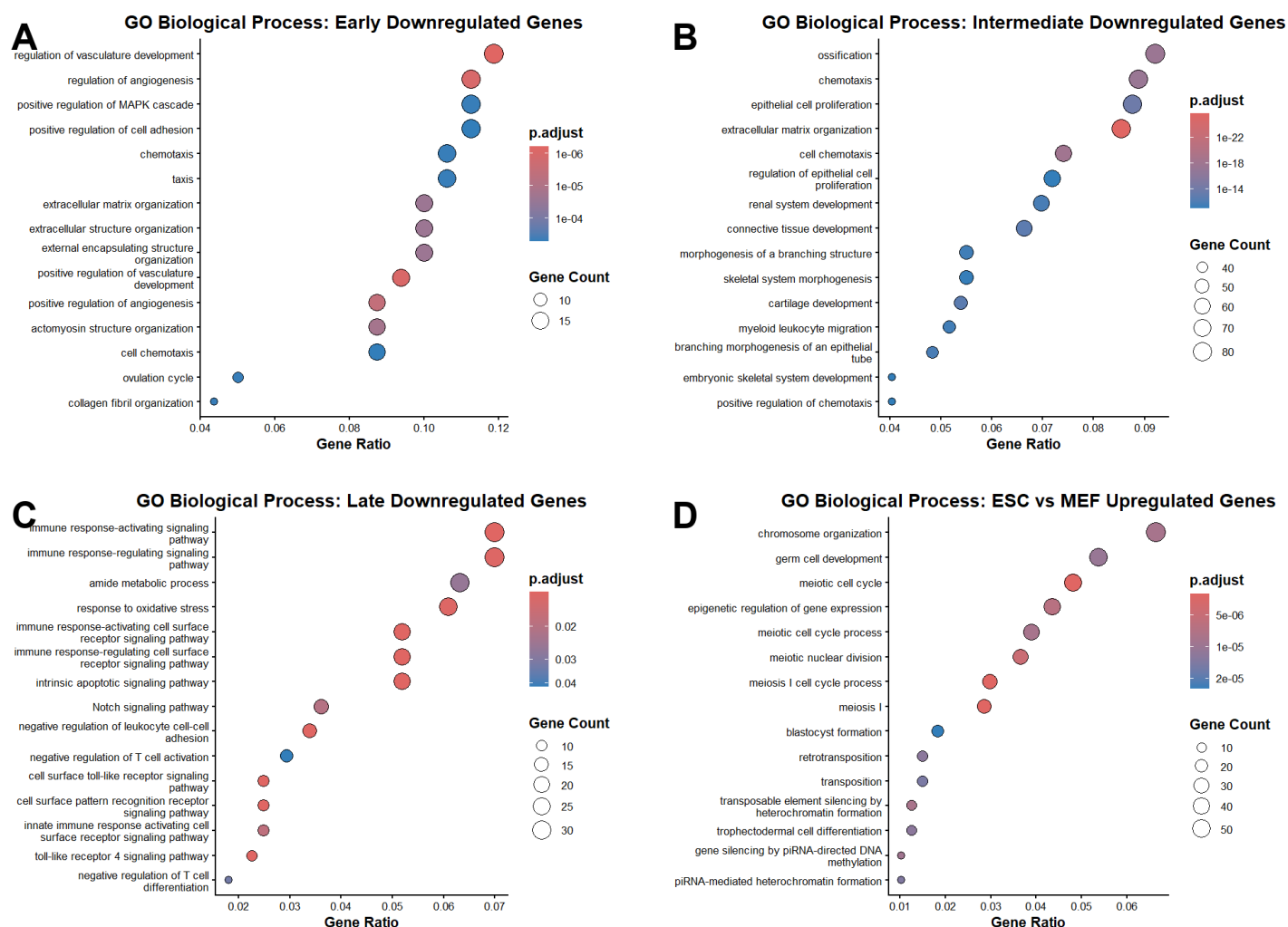
- enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*. 2023;51(D1):D638–D646. doi:10.1093/nar/gkac1000.
8. Takahashi K, Yamanaka S. Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. *Cell*. 2006;126(4):663–676. doi:10.1016/j.cell.2006.07.024.
 9. Yu G, He QY. ReactomePA: an R/Bioconductor package for reactome pathway analysis and visualization. *Molecular BioSystems*. 2016;12(2):477–479. doi:10.1039/C5MB00663E.

Supplementary material



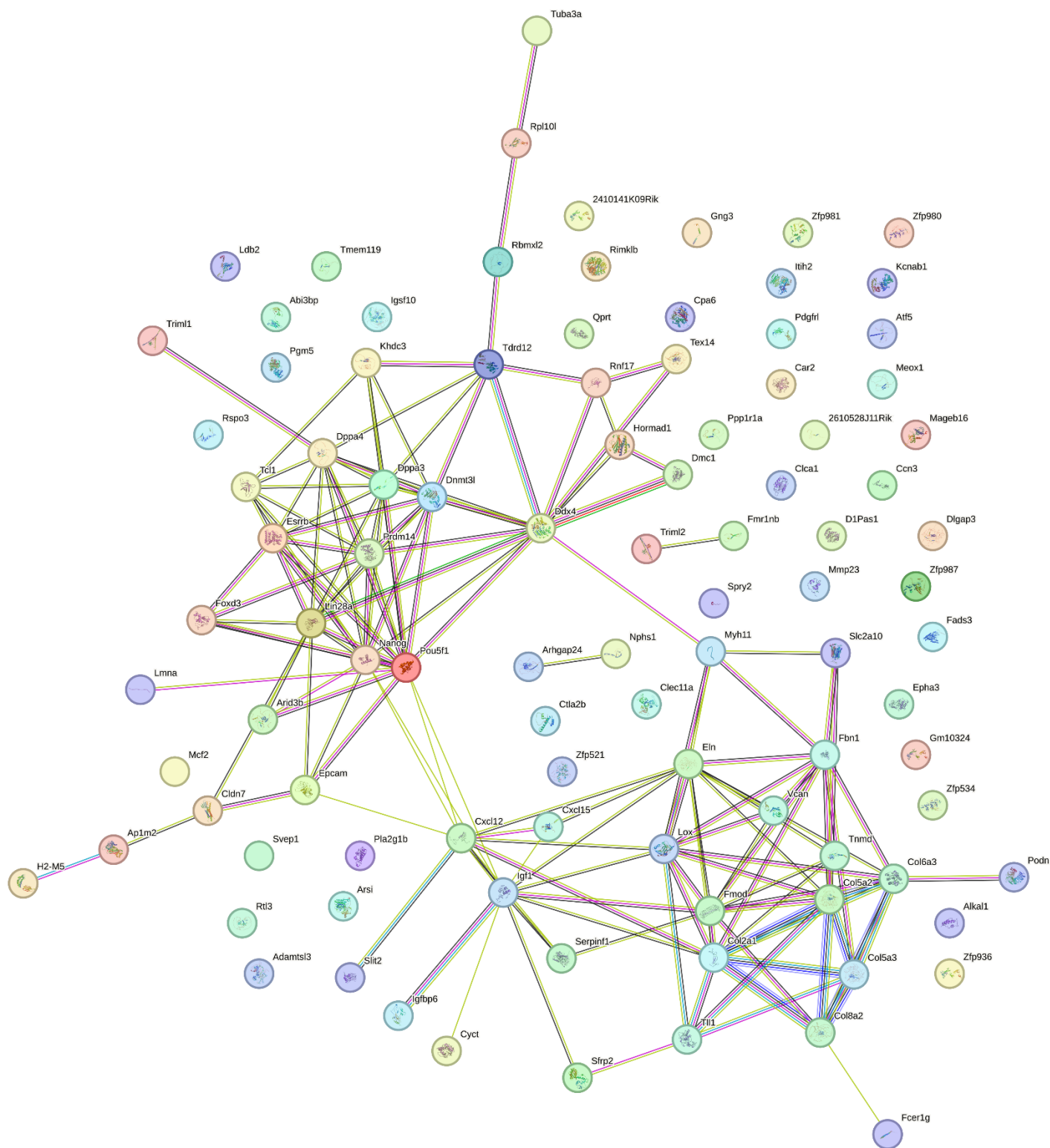
Supplementary Figure 1. Additional quality control analyses

- A* — Distribution of raw RPKM expression values before log₂ transformation.
- B* — Distribution of log₂(RPKM + 1)-transformed expression values across the analyzed samples.
- C* — Hierarchical clustering of samples based on transcriptomic expression profiles.



Supplementary Figure 2. Additional functional enrichment analyses

- A — GO Biological Process enrichment of early downregulated genes.
 B — GO Biological Process enrichment of intermediate downregulated genes.
 C — GO Biological Process enrichment of late downregulated genes.
 D — GO Biological Process enrichment of genes upregulated in ESCs relative to MEFs.



Supplementary Figure S3. STRING protein–protein interaction network centered on SPRY2.